The Cheap Yellow Display: A $15 Shack Gadget

By Ed Barnes · November 2, 2026

For about $15 you can put a live solar-weather, greyline and DX-spot display on your operating desk. The board that makes it possible is the "Cheap Yellow Display," and it may be the best-value gadget in the shack this year.

What is the CYD?

The Cheap Yellow Display (CYD) is the hobbyist nickname for the **ESP32-2432S028R**, a yellow circuit board with a 2.8-inch color touchscreen bolted on the front. Brian Lough coined the name and maintains the community reference page.

- **Brain:** dual-core ESP32 with built-in Wi-Fi and Bluetooth
- **Screen:** 2.8-inch, 320 × 240 color TFT (ILI9341) with resistive touch
- **Extras:** microSD slot, RGB LED, a few spare I/O pins
- **Power:** USB, roughly 120–180 mA at 5 V, under 1 watt

That last line matters to hams: it runs all day from a phone charger or power bank, needs no operating system or SD card, and boots in seconds. Think of it as a tiny, always-on Wi-Fi terminal that you program over USB.

Where to buy

Source	Typical price	Notes
AliExpress	$10–$15 shipped	Cheapest; 2–4 week shipping. Search "ESP32-2432S028R 2.8" and confirm the 2.8-inch version.
Makerfabs	~$15–$20	Reportedly ships with a 16 GB SD card.
Amazon	~$15–$25	Fastest delivery; often sold in 1- or 2-packs.

Two buying tips. First, newer boards have two USB ports (micro-USB and USB-C) and may show inverted colors; most firmware has a setting to fix it. Second, make sure you have a USB **data** cable, not a charge-only one. Prices are as of this issue and approximate.

Projects worth a look

The CYD community keeps a project list of ready-to-flash firmware. A few that suit a radio club:

Project	What it does
CYD Ham Dashboard	HamClock-style shack monitor: solar data, greyline map, DX cluster, POTA spots (featured below)
Aura	Weather forecast display using free Open-Meteo data
cydWeeWX	Readout for a home weather station running WeeWX
CYD-Klipper	Wireless status screen for Klipper 3D printers
CYD Stream Deck	Touch macro buttons over Bluetooth; handy for logging or digital-mode shortcuts
SmartEnergyMeter	ESPHome display for solar and battery data, useful for off-grid stations
Galagino	Classic arcade emulator, for Field Day downtime

If you write your own code, the board works with the Arduino IDE, PlatformIO, MicroPython and ESPHome.

HamClock on a CYD

Many of us run HamClock, the Raspberry Pi clock by Elwood Downey, WB0OEW, who became a Silent Key in early 2026. His data servers were due to go dark in June 2026; Pi users keep running by pointing HamClock at a community backend such as hamclock.com. The original HamClock does not run on the CYD, but the open-source **CYD Ham Dashboard** by HenrysCat brings the essentials to it, as described by 9M2PJU at hamradio.my (August 2026).

It has seven pages: clock (UTC and local), HF propagation from HamQSL (SFI, sunspots, A and K index), VHF conditions, a greyline map with your QTH, PSKReporter reception map, DX cluster spots and POTA spots.

Programming it (no coding needed)

1. Plug the CYD into your computer with a data cable.
2. In Chrome, Edge or Brave, open the web flasher at henryscat.github.io.
3. Click **Install**, pick the serial port (usually CH340 or CP2102) and confirm. Flashing takes about a minute.
4. Prefer to build from source? Clone the repo, copy `include/app_config.example.h` to `include/app_config.h`, and run `pio run -t upload` in PlatformIO.

First-time setup

1. From a phone, join the Wi-Fi network **CYD-HamClock-Setup** (password `hamclock`).
2. Browse to `http://192.168.4.1`.
3. Enter your home Wi-Fi, callsign, 6-character grid square and time zone, then save.
4. The board reboots, syncs time and fills in the data.

Operating it

- Tap the **right third** of the screen for the next page, the **left third** for the previous one.
- Tap the **center** to refresh data.
- Fine-tune settings (brightness, DX cluster telnet host, POTA distance filter, color inversion, rotation) at `http://cyd-ham.local/`.
- Hold the **BOOT** button for 5 seconds to factory-reset.

An evening's work gets you a sub-1-watt shack display that is also a good first ESP32 project for newer members.

Sources

- ESP32 Cheap Yellow Display community page (Brian Lough)
- CYD project list
- CYD Ham Dashboard article, 9M2PJU, hamradio.my
- CYD Ham Dashboard source code (HenrysCat)
- Community HamClock backend server live (Amateur Radio Daily, Feb 2026)
- Silent Key: Elwood Downey, WB0OEW (ARNewsline)